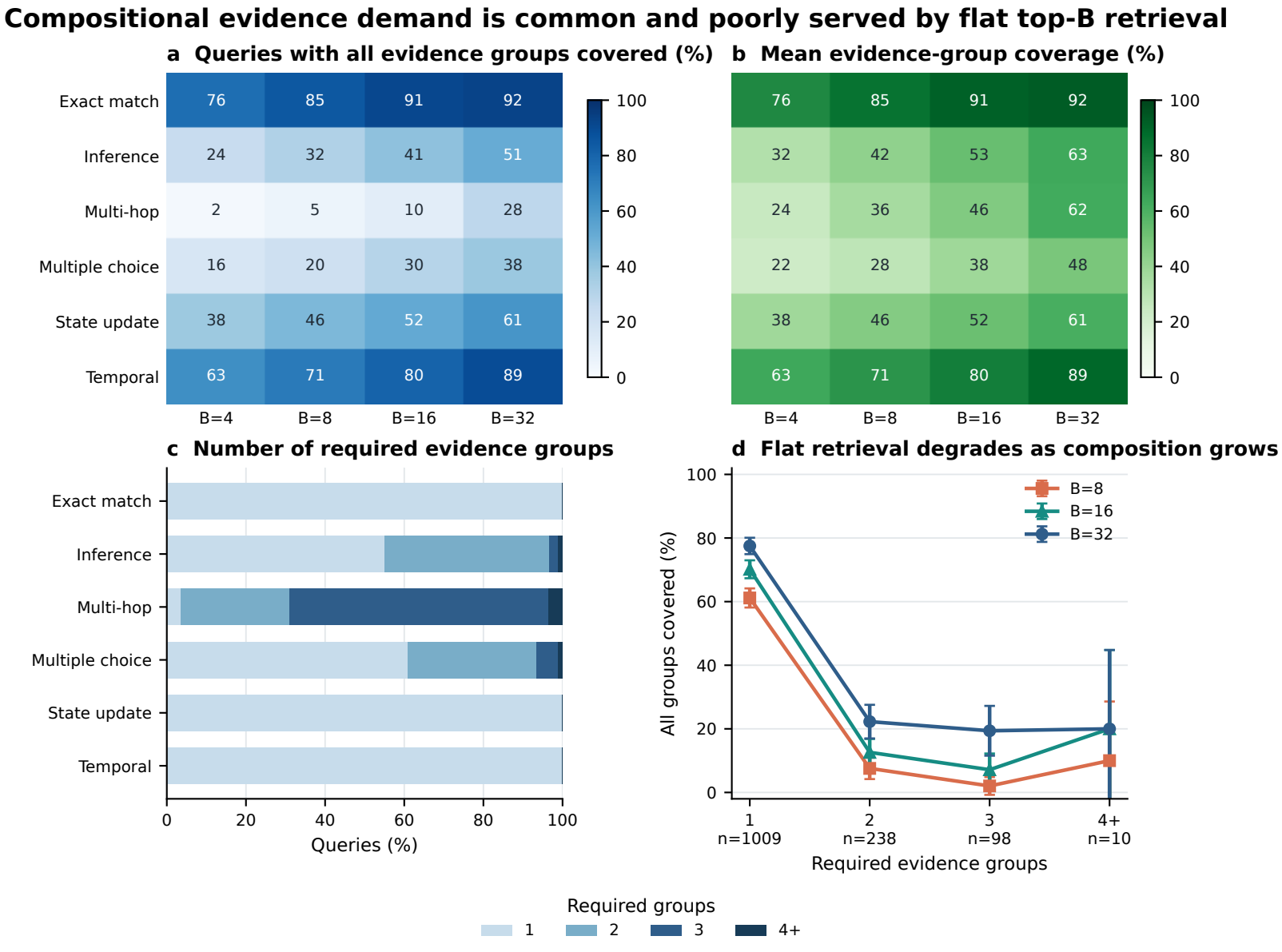
Compositional evidence demand is common and poorly served by flat top-B retrieval
a Queries with all evidence groups covered (%)
Exact match 76 85 91 92
Inference 24 32 41 51
Multi-hop 2 5 10 28
Multiple choice 16 20 30 38
State update 38 46 52 61
Temporal 63 71 80 89
B=4 B=8 B=16 B=32
100 80 60 40 20 0
b Mean evidence-group coverage (%)
76 85 91 92
32 42 53 63
24 36 46 62
22 28 38 48
38 46 52 61
63 71 80 89
B=4 B=8 B=16 B=32
100 80 60 40 20 0
c Number of required evidence groups
Exact match
Inference
Multi-hop
Multiple choice
State update
Temporal
0 20 40 60 80 100
Queries (%)
d Flat retrieval degrades as composition grows
All groups covered (%)
100 80 60 40 20 0
B=8
B=16
B=32
1 n=1009
2 n=238
3 n=98
4+ n=10
Required evidence groups
Required groups
1 2 3 4+